

Display Types -- Guided Notes ANSWER KEY

Unit tp-2.4 | CompTIA Network+ N10-009 Obj. FC0-U71 2.4 | N10-008 FC0-U71 2.4

For instructor use / distribute after students complete the notes.

Question 1

LCD (Liquid Crystal Display) monitors use a _____ to illuminate a liquid crystal panel. LEDs are used as the backlight in most modern LCDs. The liquid crystals themselves _____ (produce/block and filter) light to create the image.

Answer: Backlight; block and filter.

Question 2

OLED (Organic Light Emitting Diode) displays differ from LCDs because each pixel _____ its own light -- there is no separate backlight. This allows OLED to produce perfect _____ (true 0 luminance) by simply turning pixels off. The downside is potential _____ from prolonged static images.

Answer: Emits (produces); blacks; burn-in.

Question 3

Resolution is measured in _____ x _____ (horizontal x vertical). Common resolutions include 1920x1080 (_____ or Full HD), 2560x1440 (_____), and 3840x2160 (_____ or 4K UHD). Higher resolution displays show _____ detail but require more GPU power.

Answer: Pixels; pixels; 1080p (FHD); 1440p (QHD); 2160p (4K); more.

Question 4

Refresh rate is measured in _____ (Hz) and indicates how many times per second the display updates the image. A 60 Hz monitor updates _____ times per second. A 144 Hz gaming monitor updates _____ times per second, making fast motion appear _____ and more responsive.

Answer: Hertz; 60; 144; smoother.

Question 5

Response time is measured in _____ (ms) and indicates how quickly a pixel can change from one color to another. Lower response time reduces _____, a blurring effect seen behind fast-moving objects. For competitive gaming, a response time of _____ ms or less is generally preferred.

Answer: Milliseconds; ghosting; 1-5 ms.

Question 6

Three major display connector types are _____ (older analog standard), _____ (digital, carries audio and video, widely used), and _____ (high-speed digital, supports high refresh rates and daisy-chaining). Modern GPUs and monitors have moved away from _____ because it is an analog signal.

Answer: VGA; HDMI; DisplayPort; VGA.

Question 7

Brightness is measured in _____ (nits). A typical office monitor is around 250-350 nits. HDR (High Dynamic Range) content requires _____ peak brightness to show the difference between dark and bright areas. Viewing angle determines how far off-center a user can sit before _____ shifts.

Answer: Nits (cd/m²); high (600+ nits); color.

Question 8 -- Real World Application

REAL WORLD: A graphic designer is buying a new monitor for photo and video editing work. They are choosing between a high-refresh-rate gaming monitor (165 Hz, TN panel) and an IPS panel color-accuracy monitor (60 Hz, wide color gamut). Which should they choose and why? What is the most important spec for their use case?

Answer: The IPS color-accuracy monitor is the correct choice. For photo and video editing, the most important specifications are: color accuracy (Delta E), color gamut coverage (sRGB or Adobe RGB/DCI-P3), and panel type. IPS panels offer wide viewing angles and accurate color reproduction. TN panels, while fast, have noticeably worse color accuracy and narrow viewing angles -- colors shift when viewed off-center. A graphic designer needs to be confident that the colors they see on screen match what will print or what clients will see on other devices. High refresh rate and response time are priorities for gaming, not creative work. The 60 Hz IPS is the professional choice.